

Conditioning in Generative Quantum Denoising Diffusion Models

Daniel Quinn,^{1,*} Lorenzo Buffoni,² Stefano Gherardini,^{3,4} and Gabriele De Chiara^{1,5}

¹*Centre for Quantum Materials and Technology, School of Mathematics and Physics,
Queen's University Belfast, Belfast BT7 1NN, United Kingdom*

²*Department of Physics and Astronomy, University of Florence, 50019 Sesto Fiorentino, Italy*

³*Istituto Nazionale di Ottica del Consiglio Nazionale delle Ricerche (CNR-INO), Largo E. Fermi 6, 50125 Firenze, Italy*

⁴*European Laboratory for Non-linear Spectroscopy,
Università di Firenze, 50019 Sesto Fiorentino, Italy*

⁵*Física Teòrica: Informació i Fenòmens Quàntics, Departament de Física,
Universitat Autònoma de Barcelona, 8193 Bellaterra, Spain*

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Quantum denoising diffusion models have recently emerged as a powerful framework for generative quantum machine learning. In this work, we extend these models by introducing a conditioning mechanism that enables the generation of quantum states drawn from multiple target distributions. By sharing parameters across distinct classes of quantum states, our approach avoids the need to train separate models for each distribution. We validate our method through numerical simulations that span single-qubit generation tasks, entangled state preparation, and many-body ground state generation. Across these tasks, conditioning significantly reduced the error of targeted state generation by up to an order of magnitude. Finally, we perform an ablation study to quantify the effect of key hyperparameters on the model performance.

I. INTRODUCTION

Quantum machine learning (QML) investigates how quantum algorithms can be leveraged to learn patterns from data more efficiently than their classical counterparts [1–3]. Within this field, *generative* quantum models aim to learn and sample from quantum or classical data distributions, enabling applications in quantum simulation, quantum chemistry, and quantum state preparation [4]. Existing approaches include models such as quantum circuit Born machines [5, 6], quantum generative adversarial networks [7–10], and quantum Boltzmann machines [11–13].

More recently, *Quantum Denoising Diffusion* (QDD) models have emerged as promising candidates for quantum state generation [14, 15], inspired by the success of classical denoising diffusion probabilistic models in image creation [16–18]. QDD models implement a forward diffusion process that gradually randomises the target state or distribution, followed by a learned reverse process that removes noise and reconstructs the target state or distribution. Several variants have been proposed, including hybrid classical–quantum schemes [19–23], yet these often face difficulties in generalising to multimodal quantum data distributions.

In classical generative modelling, *conditioning* is a well-established technique that enables a single model to represent multiple target distributions, improving flexibility and reducing the need for separate models [24]. By contrast, quantum conditioning remains comparatively underexplored. Prior works have typically encoded class labels in ancilla qubits prepared in computational basis

states [14], an inherently discrete and inflexible strategy. Notably, the QDD variant of Zhang *et al.* [22], which serves as the baseline for our study, explicitly identified efficient conditioning as an open challenge.

In this work, we introduce the *Conditioned Quantum Denoising Diffusion* (CQDD) model, which extends the QDD framework by incorporating a continuous conditioning mechanism. We show that our CQDD model can successfully learn diverse quantum data distributions, including quantum many-body phases, entangled states, and states with non-trivial topological ring structure. We benchmark our CQDD against unconditioned QDD models, finding that conditioning reduces the error in targeted state generation by up to an order of magnitude. Finally, we present an ablation study of key hyperparameters to investigate the trade-offs between model complexity and generative performance.

The paper is organised as follows. In Sec. II, we describe the design of the CQDD model, including details about circuit implementations. Following this, in Sec. III, we outline the loss functions we use for training as well as the training methodology. We evaluate the CQDD model on various generation problems in Sec. IV, and highlight the need for conditioning for certain generation tasks. Finally, in Sec. V, we perform an ablation study of the model.

II. CONDITIONED QUANTUM DENOISING DIFFUSION MODEL

We consider the task of generating new quantum states drawn from an unknown distribution, given access only to finite samples. Extending the standard quantum generative modelling problem, we aim to train a single model capable of generating quantum states from multiple dis-

* dquinn53@qub.ac.uk

tributions, which is not easily achieved by models with no conditioning (see Sec. IV E). Each distribution corresponds to a different *class* or family of quantum states. Unlike traditional approaches, which typically require training a separate model for each target distribution, we seek a unified parameter set $\Theta = \{\theta_1, \dots, \theta_T\}$ that enables the model to generate samples from different distributions. This structure allows the model to selectively generate quantum states from the desired distribution based on an external conditioning input (e.g., a label).

As illustrated in Fig. 1, the CQDD model is divided into two parts, similar to the model proposed in [22]. These two parts represent opposing processes, the *forward diffusion* process and the *backward denoising* process. In the forward process, we begin with a target distribution and gradually add noise, as detailed below, to each state until the distribution becomes Haar random, which we denote as $\mathcal{H}$ [25]. This procedure is applied to each distribution that we aim to model for a total of T steps. In the backward process, a model is trained to reverse this transformation, starting from the random distribution and iteratively refining it into the desired target distribution, conditioned on an external input.

The forward diffusion process is implemented by iteratively applying scrambling circuits [26]. At each step $t = 1, \dots, T$, a unitary $U_{i,j}^{(t)}$ is applied to the n qubit quantum state $|\psi_{i,j}^{(t-1)}\rangle$, where i indexes the i^{th} sample from the distribution class j . Each unitary $U_{i,j}^{(t)}$ is constructed from randomly sampled single- and two-qubit rotations as shown in Fig. 1(c). The single qubit rotations are expressed as $RZ(\omega) = e^{-i\omega Z/2}$ which represents a rotation about the z -axis by an angle ω , with Z the Pauli- Z operator. A similar definition holds for RY and RX which are related to the Pauli- Y and X operators respectively. The two qubit rotations are given by $RZZ(\omega) = e^{-i\omega Z_p Z_q/2}$, applied between every unique pair of qubits (p, q) . A subscript on a Pauli operator indicates that it acts only on the specified qubit (e.g., Z_p acts solely on qubit p). For all these gates, rotation angles are independently drawn from the uniform distribution

$$\omega \sim \mathcal{U}\left(-\frac{\delta_t \pi}{8}, \frac{\delta_t \pi}{8}\right), \quad (1)$$

where δ_t is a schedule parameter that controls the noise strength at each diffusion step (δ_t is discussed in more detail below). Importantly, although the circuit diagram in Fig. 1(c) uses the shorthand labels RZ , RY , and RZZ , each occurrence of these rotation gates represents an independent sampling of ω .

The choice of the interval in Eq. (1), here set to $\pm\pi/8$, is arbitrary, as the noise intensity can be adjusted by tuning the schedule δ_t . However, the key requirement is that the final set of states approximate the Haar random distribution $\mathcal{H}$. This iterative process transforms each initial state $|\psi_{i,j}^{(0)}\rangle$ into a progressively more scrambled

version $|\psi_{i,j}^{(t)}\rangle$, defined by

$$|\psi_{i,j}^{(t)}\rangle = \left(\prod_{k=1}^t U_{i,j}^{(k)}\right) |\psi_{i,j}^{(0)}\rangle,$$

where the product is ordered such that earlier unitaries are applied first. The resulting set of scrambled states at step t for class j is denoted by $S_j(t) = \{|\psi_{i,j}^{(t)}\rangle\}$.

The prefactor δ_t in Eq. (1) depends on the diffusion step t and defines the *noise schedule*, analogous to its role in classical diffusion models [16]. The schedule governs how much randomness is injected at each step, with δ_t increasing over time to ensure that the applied rotations progressively explore more of the Hilbert space. This gradual increase in noise drives the set of quantum states toward a Haar random distribution [27]. If the noise schedule increases too slowly, the final state set may not fully randomise. Conversely, if it increases too rapidly, the states may become effectively random too early in the process, reducing the learnability of the reverse process.

The reverse denoising process is implemented through a sequence of parametrised quantum circuits $\tilde{U}(\theta_k)$ applied at each backward step $k = T, \dots, 1$ [5]. The bold notation indicates a vector. Explicitly, $\theta_k \in \mathbb{R}^d$ is a vector of length d consisting of real parameters $\theta_{k,d}$ for each circuit $\tilde{U}(\theta_k)$. The parameters $\theta_{k,d}$ correspond to the rotation angles of the RX and RY gates, as displayed in Fig. 1(d). Beginning with a Haar random initial state $|\tilde{\psi}_{i,j}^{(T)}\rangle$, we compose it with an ancilla register of n_a qubits prepared according to a conditioning angle μ , as detailed below. At each step, the circuit $\tilde{U}(\theta_k)$ acts jointly on the system and ancilla, and is followed by a projective measurement on the ancillary subsystem. This projective measurement realises an effective non-unitary map on the system, approximating the inverse of the forward diffusion step. The ancilla is then discarded, yielding the updated state $|\tilde{\psi}_{i,j}^{(T-1)}\rangle$. Repeating this process across all T reverse steps and for all states yields the reconstructed set at step t , denoted by $\tilde{S}_j(t) = \{|\tilde{\psi}_{i,j}^{(t)}\rangle\}$ for a class j .

Each denoising unitary is implemented as an ansatz with L layers of parametrised single-qubit rotations with entangling gates, following the architecture of Ref. [28]. The entangling gates are controlled- Z operations, $CZ = \text{diag}(1, 1, 1, -1)$, applied to nearest-neighbour qubits in an alternating pattern. The full circuit is shown in Fig. 1(d), where the ancilla qubits are initialised in the state $(RX(\mu)|0\rangle)^{\otimes n_a}$. The conditioning angle μ is defined in the range $0 \leq \mu < 2\pi$. For a task consisting of $|\mathcal{C}|$ classes, we take $|\mathcal{C}|$ linearly spaced values of μ in this range. For example, if $|\mathcal{C}| = 2$ then we assign $\mu = 0$ and $\mu = \pi$ to the different classes, or if $|\mathcal{C}| = 3$, we assign $\mu = 0, \pi/3$ and $2\pi/3$. The total number of parameters in a single $\tilde{U}(\theta_k)$ is $d = 2L(n + n_a)$.

This continuous conditioning mechanism allows the model to encode more labels than allowed from simply initialising the ancilla in the orthogonal computational basis states (e.g. $|00\rangle, |01\rangle, |10\rangle, |11\rangle$), under the

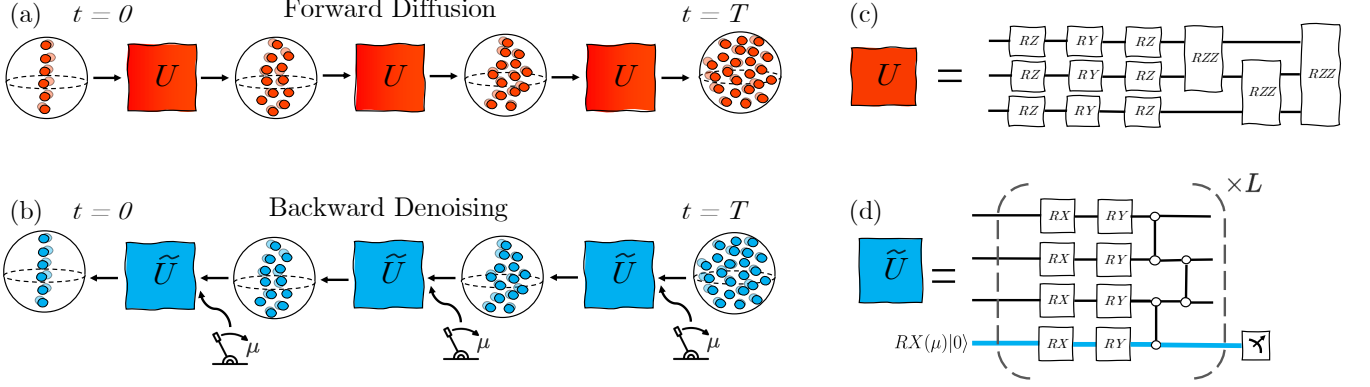


FIG. 1. **Model Schematic.** (a) The forward diffusion process begins with a structured state distribution $S_j(0)$ (left most Bloch sphere), here in the figure chosen as states distributed along a ring (red dots). Each state is passed through a scrambling unitary $U = U_{i,j}^{(t)}$ (red boxes), and repeated applications of the latter yield a Haar random distribution $S_j(T)$ (right most Bloch sphere). (b) The reverse denoising process starts from $\tilde{S}_j(T)$ and attempts to reconstruct the original ring distribution. Each state passes through a parametrised unitary $\tilde{U} = \tilde{U}(\theta_t)$ that comprises ancilla qubits, measured and discarded at each backward step. The initial state of the ancilla qubits are conditioned on an angle μ (represented by the lever). Depending on the number of classes to generate from, we take μ to be linearly spaced values in the range $0 \leq \mu < 2\pi$. (c) Scrambling circuits use random rotation angles sampled according to Eq. (1) for an $n = 3$ qubit state. (d) The denoising circuits (example with $n = 3$, $n_a = 1$) is implemented using an ansatz of L layers of single-qubit rotations and entangling gates. In this circuit, measurements act only on the ancilla qubits initialised as $(RX(\mu)|0\rangle)^{\otimes n_a}$, which is illustrated by the thick blue line. The remaining qubits encode the system state $|\tilde{\psi}_{i,j}^{(t)}\rangle$.

assumption to consider a fixed number of ancilla qubits n_a . Throughout this manuscript, we use $n_a = 2$, unless stated otherwise, as we found it to be optimal. However, in practice, n_a is another hyperparameter to tune.

In the following, we benchmark our rotation-based conditioning strategy against a discrete approach in which each class is assigned a distinct computational basis state of the ancilla register (see Appendix A). Remarkably, we find that rotation-based conditioning yields a lower generative error. This suggests that encoding class information via continuous rotations allows the model to capture class structure more effectively, enhancing the performance of the model at the testing stage with new samples from the initial Haar random distribution $\mathcal{H}$.

III. LOSS FUNCTION AND TRAINING METHODOLOGY

To allow for training of the model, one first needs a loss function to minimise, i.e., we require a metric that quantifies the distance between two sets A and B of states. Following the approach in Ref. [22], suppose the two sets are finite with $A = \{|\phi_i\rangle\}_{i=1}^N$ and similarly $B = \{|\psi_i\rangle\}_{i=1}^N$ where N is the number of states. Then, we define the pairwise fidelity

$$\bar{F}(A, B) = \mathbb{E}_{|\phi\rangle \in A, |\psi\rangle \in B} [F(\psi, \phi)], \quad (2)$$

which is based on the simple statewise fidelity $F(\psi, \phi) = |\langle \psi | \phi \rangle|^2$. Using the pairwise fidelity (2), we build the so-called maximum mean discrepancy (MMD) distance [29]. The MMD distance between the sets can be calculated as

$$\mathcal{D}_{\text{MMD}}(A, B) = \bar{F}(A, A) + \bar{F}(B, B) - 2\bar{F}(A, B). \quad (3)$$

Alternatively, one can use the Wasserstein distance for when it is infeasible to distinguish between two state sets with the MMD distance. Such scenarios show up when one has a distribution that forms a ring, as shown in the Supplementary Material of Ref. [22]. The Wasserstein distance is computed by casting it into a linear optimisation problem [30], which we are going to briefly recall. Suppose, as before, that $A = \{|\phi_i\rangle\}_{i=1}^N$ and similarly $B = \{|\psi_i\rangle\}_{i=1}^N$. Then, we define a $N \times N$ cost matrix C , with entries given by the infidelity $C_{i,j} = 1 - |\langle \psi_i | \phi_j \rangle|^2$. To compute the Wasserstein distance, we also require a probability distribution governing the sampling of $|\phi_i\rangle \sim A$, and likewise $|\psi_i\rangle \sim B$. Here, as we wish to generate any state from these distributions, we set the probability distributions to be uniform. Hence, we define the Wasserstein distance as the solution to the following linear program:

$$\mathcal{D}_{\text{WASS}}(A, B) = \min_P \sum_{i,j} P_{ij} C_{ij}, \quad (4)$$

subject to $\sum_j P_{ij} = \sum_i P_{ij} = N^{-1}$ and $P_{ij} \geq 0 \quad \forall i, j$.

Using the MMD or Wasserstein distance, the loss function is defined as the sum of the distances computed for each class within the dataset. Mathematically, this is expressed as

$$\mathcal{L}(t) = \frac{1}{|\mathcal{C}|} \sum_{j=1}^{|\mathcal{C}|} \mathcal{D}(S_j(t), \tilde{S}_j(t)), \quad (5)$$

with $|\mathcal{C}|$ denoting the number of classes. This is a relatively simple loss function as we assume all classes have the same number of states, i.e. $|S_j(t)| = |\tilde{S}_j(t)| = N$ for $j = 1, \dots, |\mathcal{C}|$ and all steps t . If, for instance, this was not the case one could employ a weighted average to counteract the bias of having an unequal number of data. Additionally, we use a normalised version of the Wasserstein (or MMD) distance, where the distance is divided by $\max_j \mathcal{D}_{\text{WASS}}(\mathcal{H}, S_j(0))$. This normalisation accounts for the fact that the distance between the target set $S_j(0)$ and the Haar distribution $\mathcal{H}$ can vary significantly across problem settings, for instance, due to differing n . As a result, the value $\mathcal{L}(t)$ of the loss function can be interpreted as a relative error, expressed as a percentage with 100% meaning completely Haar random.

The training methodology of the model follows a *divide-and-conquer* approach. We begin by optimising the parameters of the first backward step, θ_T , to minimise the loss function $\mathcal{L}(T)$. This optimisation is performed using the Adam optimiser with a fixed learning rate [31]. The parameters θ_T are initialised according to a standard Gaussian distribution, $\mathcal{N}(0, 1)$. With these initial values, we generate the corresponding output states from $\tilde{U}(\theta_T)$ and compute the loss $\mathcal{L}(T)$. The Adam optimiser is then applied to update θ_T , and after 5000 iterations the optimisation is halted, we fix θ_T to the values that yield the minimum for $\mathcal{L}(T)$.

We then proceed to the denoising circuit $\tilde{U}(\theta_{T-1})$ and compute the loss $\mathcal{L}(T-1)$, again applying the same optimisation procedure. This process is repeated iteratively by progressing backward through time steps until $t = 0$, so as to obtain the optimal set of parameters $\Theta = \{\theta_1, \dots, \theta_T\}$. In essence, this training procedure consists of T sequential optimisation problems, each dependent on the parameters learnt in all subsequent steps.

IV. NUMERICAL SIMULATIONS

In this section, we evaluate the CQDD model on four diverse datasets to showcase its flexibility. In particular, we consider: single-qubit states arranged along a ring on the Bloch sphere, clusters of states centred around six poles of the Bloch sphere, Bell states with random relative phases, and ground states of the transverse-longitudinal field Ising model. More importantly, we also evaluate the performance of the conditioned model against an equivalent unconditioned model, and demonstrate the advantage that our model has for generative tasks. Similar examples for single class generation (i.e. no

conditioning) are shown in Ref. [22]. Detailed hyperparameters used in our models for each of these tasks are provided in Appendix E.

A. Topological Structure

In the case of a single qubit ($n = 1$), we define three classes of quantum states that the CQDD model aims to regenerate. These classes are defined as follows:

$$S_X = \{\cos(\phi)|0\rangle - i\sin(\phi)|1\rangle \quad \text{s.t.} \quad \phi \sim \mathcal{U}(0, 2\pi)\}, \quad (6)$$

$$S_Y = \{\cos(\phi)|0\rangle + \sin(\phi)|1\rangle \quad \text{s.t.} \quad \phi \sim \mathcal{U}(0, 2\pi)\}, \quad (7)$$

$$S_Z = \left\{(|0\rangle + e^{-i\phi}|1\rangle) / \sqrt{2} \quad \text{s.t.} \quad \phi \sim \mathcal{U}(0, 2\pi)\right\}. \quad (8)$$

Geometrically, S_X corresponds to a ring lying in the Y -plane of the Bloch sphere. Similarly, S_Y and S_Z correspond to rings in the XZ - and XY -planes, respectively.

Each of these classes undergoes the forward diffusion process using the same noise schedule, where a fixed diffusion factor δ_t is applied uniformly at every step. Our model conditions the reverse denoising process using a conditioning angle μ assigned arbitrarily to each ring: S_X is assigned to $\mu = 0$, S_Y to $\mu = 2\pi/3$, and S_Z to $\mu = 4\pi/3$. As a performance metric, we consider the Wasserstein distance

$$\mathcal{D}_{\text{WASS}}(\tilde{S}_j(t), S_j(0)), \quad j \in \{X, Y, Z\}, \quad (9)$$

as shown in Fig. 2(a)-(c). This distance measures how far the diffused or generated set at step t deviates from its corresponding initial set. In both training and testing, we obtain a comparable final loss of $\mathcal{L}_{\text{WASS}}(0) \approx 4.3\%$. To illustrate the fidelity of the learned samples, we visualise the final quantum states on the Bloch sphere in Fig. 2(d)-(e). Overall, the model performs well in the generation task, i.e., the results collectively exhibit the desired ring structure across all three distributions, and not just a single one.

B. Polar Points

In addition to the planar rings, we also consider a class of $n = 1$ qubit states that we refer to as polar points. These states are selected from the six poles of the Bloch sphere along the Cartesian axes, corresponding to the eigenstates of the Pauli operators X , Y , and Z . Explicitly, the states are given by:

$$\mathcal{C} = \left\{ |0\rangle, |1\rangle, |+\rangle, |-\rangle, |+i\rangle, |-i\rangle \right\}, \quad (10)$$

where $|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$ and $|\pm i\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm i|1\rangle)$. As we require a set of states, we add Gaussian noise to these eigenstates such that they form clusters around the

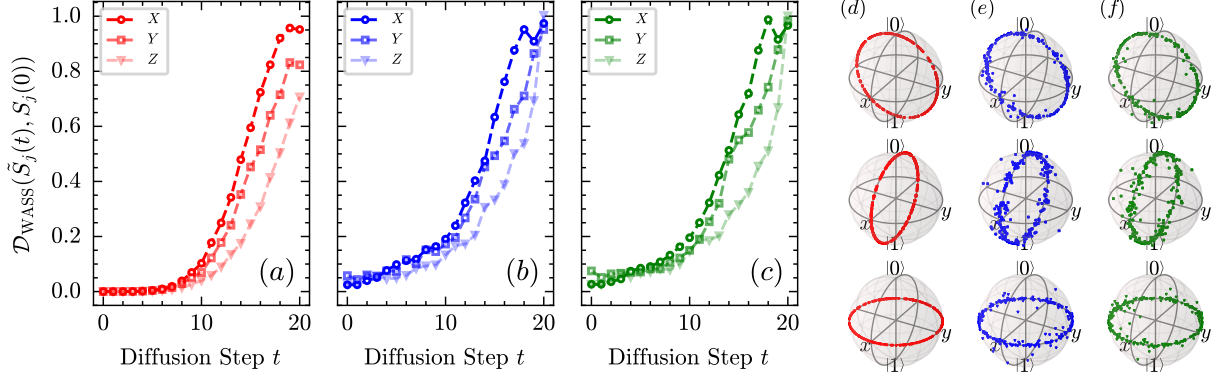


FIG. 2. **Topological Structure Results.** Training and testing performance across all three planar ring classes. The diffusion panel (a) reads left to right, while training (b) and testing (c) panels are read right to left, indicating the backwards processes. The transparency and marker shape denote S_X (dots), S_Y (squares), and S_Z (triangles). In (d)-(f) we depict the Bloch spheres for output states at diffusion step $t = 0$. Each row corresponds to S_X , S_Y , and S_Z , and each column to diffusion (red), training (blue), and testing (green). To save on rendering, only the first 250 states are plotted.

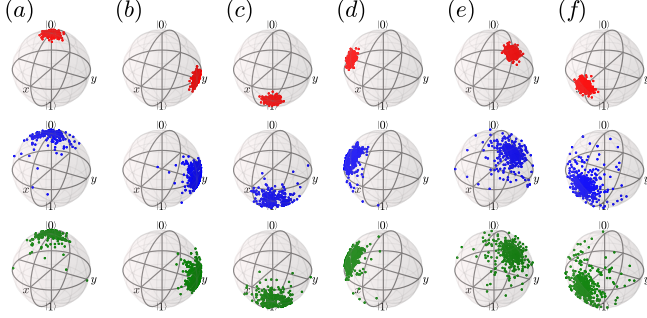


FIG. 3. **Polar Points Results.** Bloch spheres (a)-(f) show the final output states at diffusion step $t = 0$. Each row corresponds to different data sets: diffusion (red), training (blue), and testing (green). Each column represents one of the six cardinal directions (poles) on the Bloch sphere.

target directions. That is, for the $|0\rangle$ -class, these states are of the form $|\psi\rangle \sim |0\rangle + \epsilon c|1\rangle$, normalised appropriately, where $\text{Re}(c)$ and $\text{Im}(c)$ are independent and follow the Gaussian distribution $\mathcal{N}(0, 1)$. Similar states are sampled for the other classes, such as for the $|+\rangle$ -class we have $|\psi\rangle \sim |+\rangle + \epsilon c|-\rangle$, or for the $|-i\rangle$ -class we set $|\psi\rangle \sim |-i\rangle + \epsilon c|i\rangle$. The scale factor is set to $\epsilon = 0.08$ and it determines how far the states deviate from the eigenstates. The values of $3\mu = 0, \pi, 2\pi, 3\pi, 4\pi, 5\pi$, are set to correspond to the eigenstates of $+Z, +Y, -Z, -Y, +X$, and $-X$, respectively.

As we now have six total classes, this configuration tests the model's capacity to correctly generate from a larger number of classes than the discrete basis conditioning method. The limitation arises because with two ancilla qubits ($n_a = 2$) there are only $2^{n_a} = 4$ computational basis states available, so the method cannot uniquely represent more than four classes. Illustrated in Fig. 3 is the final Bloch spheres achieved at $t = 0$

Direction	Train (%)	Test (%)
+Z	1.7	0.8
+Y	1.8	2.0
-Z	1.9	2.1
-Y	1.4	1.5
+X	3.7	4.7
-X	5.2	6.4

TABLE I. **Distance of Polar Points.** We display values of $\mathcal{D}_{\text{MMD}}(\tilde{S}_j(0), S_j(0)) / \mathcal{D}_{\text{MMD}}(\mathcal{H}, S_j(0))$ for all the directions in percentages. Lower values indicate better agreement between the target distribution class and the training and testing class.

for diffusion, training and testing. Note that our model achieves a final test error of $\mathcal{L}_{\text{MMD}}(0) \approx 1.5\%$.

An alternative figure of merit is the distance of each generated class from its target, analogous to the metric in Eq. (9) but only with the MMD distance. Here, we consider the final step of the denoising process, $t = 0$, and evaluate the spread of each Cartesian axes $j \in \{+Z, +Y, -Z, -Y, +X, -X\}$ for the training and testing sets. The normalised results are reported in Table I as percentages.

From Table I, the model achieves the smallest spread for the $+Z$ and $-Y$ classes, while the $\pm X$ classes exhibit larger distances, indicating a broader distribution of generated states. This trend is also apparent in Fig. 3, where the training and testing Bloch spheres (a) and (d) are more tightly clustered than (e) and (f). We conjecture that the bias towards $+Z$ and $-Y$ is due to the corresponding conditioning angles being $\mu = 0$ and $\mu = \pi$, which appear to be more favourable for the model's training dynamics.

C. Entanglement Structure

Let us now consider generating states with more qubits ($n = 2$), where a natural choice is given by the Bell states [32]. To construct a class of entangled states, we introduce a relative phase into the standard Bell states. Specifically, we define two classes:

$$S_\Phi = \left\{ \frac{|00\rangle + e^{i\phi}|11\rangle}{\sqrt{2}} \quad \text{s.t.} \quad \phi \sim \mathcal{U}(0, 2\pi) \right\}, \quad (11)$$

$$S_\Psi = \left\{ \frac{|01\rangle + e^{i\phi}|10\rangle}{\sqrt{2}} \quad \text{s.t.} \quad \phi \sim \mathcal{U}(0, 2\pi) \right\}. \quad (12)$$

Each class is identified by a conditioning angle μ , where $\mu = 0$ corresponds to S_Φ and $\mu = \pi$ to S_Ψ . Importantly, these two classes lie in orthogonal subspaces of the Hilbert space, and, unlike the single-qubit case, this two-qubit task requires learning entanglement and correctly switching between distinct subspaces. After training our model, we obtain a final test error of $\mathcal{L}_{\text{WASS}}(0) \approx 2\%$.

As an alternative metric, the degree of entanglement within these sets of states can be quantified via the average Meyer-Wallach measure $\langle Q \rangle$. For a pure normalised state, this measure is defined as

$$Q = \frac{2}{n} \sum_{i=1}^n (1 - \text{Tr}[\rho_i^2]), \quad (13)$$

where ρ_i is the reduced density matrix for qubit i , obtained by tracing out all other qubits [33, 34]. Our model achieves a value of $\langle Q \rangle = 0.97$ when the true value is 1 in both distributions as a relative phase does not affect the entanglement measure.

We can also consider a metric that quantifies the *percentage* of a generated state that lies within a specific target subspace. For a normalised two-qubit pure state

$$|\psi\rangle = c_0|00\rangle + c_1|01\rangle + c_2|10\rangle + c_3|11\rangle,$$

with $c_i \in \mathbb{C}$, we define the percentage overlap with the $\{|00\rangle, |11\rangle\}$ subspace as $|c_0|^2 + |c_3|^2$, and similarly, the percentage overlap with the $\{|01\rangle, |10\rangle\}$ subspace as $|c_1|^2 + |c_2|^2$. In both cases, we find that any given generated state has an overlap of 99.9% with its respective subspace.

In addition, we investigate the task of generating entangled $n = 3$ qubit states belonging either to the GHZ- or to the W-class [35]. Specifically, we define two target distributions to be a GHZ-class consisting of states of the form $|\text{GHZ}\rangle = (|000\rangle + e^{i\phi}|111\rangle)/\sqrt{2}$, and a W-class that includes states of the form $|\text{W}\rangle = (|001\rangle + |010\rangle + e^{i\phi}|100\rangle)/\sqrt{3}$, with ϕ representing an arbitrary phase. Using a subspace-fidelity metric similar to that introduced earlier, we find that the model successfully generates both GHZ and W states, with approximately 90% of samples lying within their respective subspaces. However, in contrast to previous experiments, the generated distributions in the Hilbert space do not form well-structured rings. This is reflected in a relatively high

final Wasserstein loss of $\mathcal{L}_{\text{WASS}}(0) = 15\%$. We hypothesise that this larger error may stem from the difference in rank between the two target classes as GHZ states span a rank-2 subspace, whereas W states span a rank-3 subspace. Further visualisations and analysis are provided in Appendices B and C.

D. Many-body Phases

As a physically relevant example, we consider the transverse-longitudinal field Ising model (TLFIM), whose ground states have been studied extensively [36]. The model is defined by the Hamiltonian

$$H = - \sum_{i=1}^n Z_i Z_{i+1} - g \sum_{i=1}^n X_i - h \sum_{i=1}^n Z_i, \quad (14)$$

where periodic boundary conditions are imposed by identifying $Z_{n+1} \equiv Z_1$. The transverse magnetic field strength is denoted by g , and h represents the longitudinal field. In the absence of the longitudinal field ($h = 0$), the system undergoes a quantum phase transition at $g = 1$. For $g < 1$ the system is in a ferromagnetic phase and when $g > 1$ the system enters a paramagnetic phase.

For the state generation task, we focus on $n = 4$ qubit ground states drawn from two distinct quantum phases. The transverse field strength is sampled as $g \sim \mathcal{N}(0.5, 0.1)$, a Gaussian distribution with mean 0.5 and standard deviation 0.1. Two classes are constructed, with one setting taking the longitudinal field $h = 0.25$ and the another setting $h = -0.25$, denoted by S_+ and S_- respectively. These classes correspond to states with an incomplete spin alignment in the positive or negative z -direction. Furthermore, we set the conditioning angle to $\mu = 0$ for S_+ and $\mu = \pi$ for S_- . We note that decreasing $|h|$ increases the fidelity between states in S_+ and S_- , effectively reducing their separation in the Hilbert space. This may render the generation task easier, as the model no longer needs to reproduce states lying at nearly opposite regions in the Hilbert space.

To evaluate the fidelity of the generated states, we use the z -magnetisation, defined as $M = \sum_{i=1}^n Z_i$. Portrayed in Fig. 4 is the probability distribution $p(M)$ for the magnetisation in both generated and target datasets. A close alignment between the model-generated distributions and the true data is observed, while the diffused states at the end of the forward process are more dispersed, approximating a Gaussian distribution. As another metric, we report the mean magnetisation: the true data yields $\langle M_{\text{targ}} \rangle = 0.97$, the fully diffused set gives $\langle M_{\text{rand}} \rangle = 0.43$, and the model outputs achieve $\langle M_{\text{model}} \rangle = 0.88$ for both considered classes in training and testing. Additionally, the final test error of the model is $\mathcal{L}_{\text{MMD}}(0) \approx 3.7\%$.

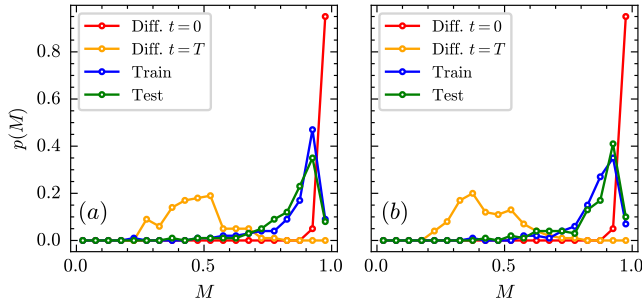


FIG. 4. **Many-body Phase Results.** Probability distributions of magnetisation for generated data from the training set (blue) and testing set (green), compared to the true data distribution (red) and the scrambled distribution at $t = T$ (orange). Panel (a) corresponds to $h = 0.25$, and panel (b) to $h = -0.25$. The shorthand Diff. $t = k$ stands for the set of data at the diffusion step k .

E. Benchmarking of Conditioned and Unconditioned Models

We now highlight a key result of our study, namely the benchmarking of conditioned versus unconditioned QDD models. Benchmarking here serves a dual role, not only as a quantitative performance test, but also as a clear demonstration of the advantage of conditioning in generative quantum modelling.

We first consider a simple $n = 1$ qubit task involving the ring structures from Section IV A, where the target distribution is $S = S_X \cup S_Y$ as defined in Eqs. (6) and (7). Using hyperparameters $T = 20$, $n_a = 2$, $L = 12$, and a total of $N = 250$ unlabelled training states, the unconditioned model reaches a final test loss of 84%, indicating poor reconstruction. In contrast, a CQDD model trained with $N = 125$ samples per class (also totalling 250) attains a loss of 7.9%, which is more than an order-of-magnitude improvement. Here, $\mu = 0$ generates the S_X class and $\mu = \pi$ generates the S_Y class. The clear difference between loss values illustrates the central advantage of conditioning. Specifically, it enables sampling from either component distribution, rather than a blurred approximation to their union.

We next address a more demanding $n = 2$ qubit task of generating maximally entangled states from the union $S = S_\Phi \cup S_\Psi$, defined in Eqs. (11) and (12). With the same hyperparameters, the unconditioned model attains a final test loss of 5.9%, whereas our CQDD model reduces this to 1.4%. Again, $\mu = 0$ targets S_Φ and $\mu = \pi$ targets S_Ψ . These results demonstrate that conditioning systematically improves generative performance.

In Fig. 5, we show the training loss of both conditioned and unconditioned models for the (a) ring task and (b) entanglement task as described above. The loss for the unconditioned model is simply $\mathcal{L}(t) = \mathcal{D}(\tilde{S}(t), S(t))$, while for the conditioned case we have $2\mathcal{L}(t) = \mathcal{D}(\tilde{S}_i(t), S_i(t)) + \mathcal{D}(\tilde{S}_j(t), S_j(t))$ where $i = X, \Phi$

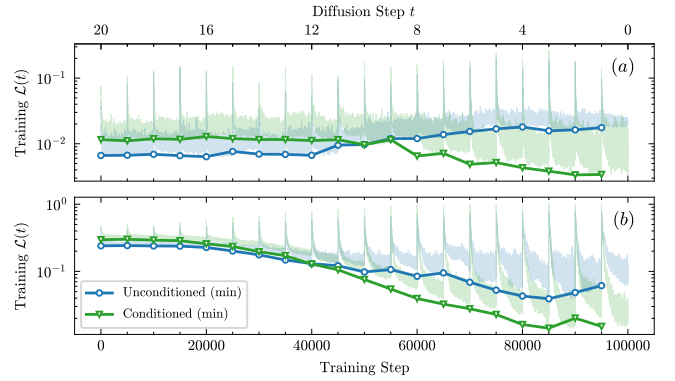


FIG. 5. **Benchmarking.** Normalised training loss for the (a) ring and (b) entanglement generation task. We compare the performance of the unconditioned model (blue) and the conditioned (green) model across training steps. Markers show the minimum loss at each diffusion step t , while the faint lines show the raw loss values during the training. Periodic spikes are caused by random initialisation at the start of every step.

and $j = Y, \Psi$ for the respective tasks. Markers in the figure denote the minimum value of the loss for a particular backwards step t , which is the value we use to find θ_t . In the faint colours we plot the raw loss value, where the periodic spikes are caused by the randomised initialisation at each backward step.

One can see a significant reduction in the final loss for the conditioned model. In the first half of the training process, we see that the value of the unconditioned loss is smaller than the conditioned in both tasks. However, the difference between the training trajectories becomes more pronounced after the crossover at approximately $T/2$, in which the loss of the conditioned model decreases rapidly compared to the unconditioned model. In fact, for the rings task in Fig. 5(a), the loss of the unconditioned model increases after $T/2$ steps, implying that the training trajectory does not converge.

V. ABLATION STUDY

In this section, we analyse how the model error scales with changes to various hyperparameters. We divide this analysis into two categories: (i) model-dependent scaling, where we vary hyperparameters such as the number of diffusion steps T , circuit depth L , and number of ancilla qubits n_a ; and (ii) problem-dependent scaling, where we vary the complexity of the generative task.

A. Model Dependent Scaling

We analyse the scaling behaviour of our model, focusing on how its performance varies with the number of data points and the number of layers in the variational circuit. As a benchmark task, we consider the three-class

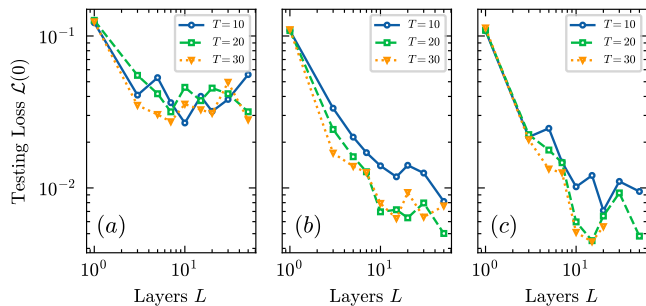


FIG. 6. **Model Scaling.** Testing loss as a function of the number L of layers. We evaluate the impact of the number of diffusion steps T by plotting the results for $T = 10$ (blue circles), $T = 20$ (green squares), and $T = 30$ (yellow triangles). Panel (a) corresponds to $N = 100$ data points, (b) to $N = 500$, and (c) to $N = 1000$.

planar ring dataset, introduced in Sec. IV A. The model’s testing loss $\mathcal{L}(0)$ is illustrated in Fig. 6 as a function of the circuit depth and is evaluated across three different dataset sizes. We further examine how performance varies with the number of diffusion steps. The results demonstrate that increasing the number of circuit layers L generally improves the test performance, which is consistent with both classical and quantum learning theory [37]. However, beyond a certain depth, the test loss saturates or begins to increase that is an indication of overfitting, meaning that the model fits the training data too closely but fails to generalise [38]. This effect is more pronounced for small datasets ($N = 100$), where overfitting occurs earlier. For larger datasets ($N = 500, 1000$), deeper circuits remain beneficial for longer. Similarly, the number of diffusion steps T influences the performance of the models. Larger T enables finer denoising and improves performance, but also introduces more parameters and thus the possibility for overfitting.

Another important scaling behaviour worth investigating is the relationship between the model performance and the number of ancilla qubits. To explore this, we consider the learning task of the Ising model introduced in Sec. IV D. Illustrated in Fig. 7 is the final normalised testing error as a function of the number of ancilla qubits under two distinct resource regimes. In the *unconstrained* regime, we fix the circuit depth to $L = 12$ and vary n_a , effectively increasing the total number of trainable parameters. In contrast, in the *constrained* regime, we fix the total number of parameters by keeping the product $Ln_a = 12$ constant, adjusting the depth inversely with the ancilla count. The model was evaluated on $N = 250$ test states, partitioned into five subsets of 50 samples each. For each subset, we compute the average test loss across both classes. The reported loss is the mean over the partitions, and the error bars indicate the standard deviation of the subsets.

As shown in Fig. 7, the two regimes have a clear distinction. In the unconstrained case, increasing n_a ini-

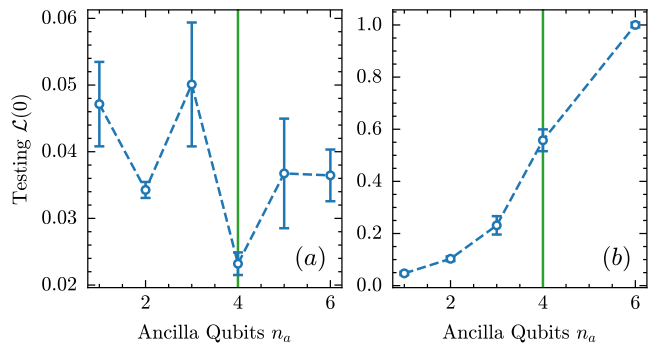


FIG. 7. **Ancilla Scaling.** Final testing loss $\mathcal{L}(0)$ as a function of the number n_a of ancilla qubits for (a) the unconstrained resource regime and (b) the fixed resource regime with constant Ln_a . In both plots, the green solid lines indicate $n = n_a$.

tially reduces the testing error, due to an increase in the parameter count. However, error eventually fluctuates, reaching a minimum at $n_a = n = 4$ (marked by the green solid line in both panels of the figure). Interestingly, we see that when $n_a \leq n$ the odd numbers of ancilla qubits ($n_a = 1, 3$) have higher losses and wider error bars, whereas the even numbered ancilla count ($n_a = 2, 4$) is much lower and fixed. We conjecture that this is due to the entanglement layers in the backwards circuit, as this is where the interplay between the ancilla and system comes into effect. However, a more thorough analysis is needed to determine why this insight breaks down for $n_a > n$, but this goes beyond the scope of this work.

In contrast, the constrained regime shows a clear monotonic increase in the error as n_a increases and the circuit depth decreases. The worst performance occurs at $n_a = 6$ and $L = 2$, highlighting that for a fixed parameter budget, deeper circuits (larger L , smaller n_a) are empirically more effective than wider, shallower ones. This aligns with classical deep learning findings, where deep models perform better than wide models [39, 40].

B. Problem Dependent Scaling

In this section, we investigate how the performance of the CQDD model scales with the problem complexity, specifically in terms of the number of target classes $|\mathcal{C}|$ to be generated. To this end, we consider two representative examples: (i) distributions of single-qubit states that form rings, which are parallel to the equator of the Bloch sphere, and (ii) various 4-qubit GHZ entangled states with varying relative phases. These tasks exemplify low- and high-complexity generative challenges, respectively.

The task *rings*, illustrated in Fig. 8, consists of generating single-qubit states drawn from the set

$$S(\alpha) = \left\{ \cos\left(\frac{\alpha}{2}\right) |0\rangle + e^{i\phi} \sin\left(\frac{\alpha}{2}\right) |1\rangle \text{ s.t. } \phi \sim \mathcal{U}(0, 2\pi) \right\} \quad (15)$$

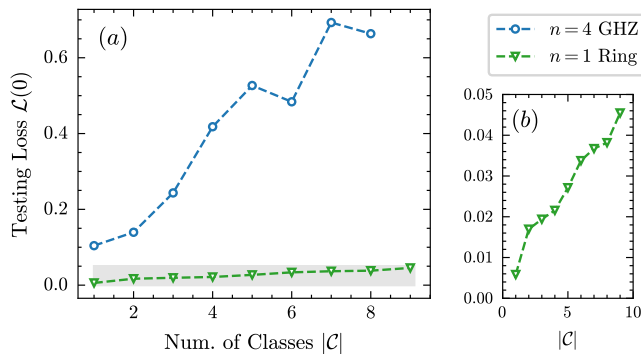


FIG. 8. **Class Scaling.** (a) Test loss $\mathcal{L}(0)$ versus the number of classes $|C|$ for the GHZ (blue circles) and rings (green triangles) tasks. (b) Enlarged view of the shaded region in (a). For both tasks, we set the following hyperparameters: $T = 20$, $L = 15$, $n_a = 2$ and $N = 100$.

where each class is associated with a fixed angle α . For a problem with $|C|$ classes, the class-dependent angles are defined as

$$\alpha = \frac{j\pi}{|C| + 1}, \quad j = 1, \dots, |C|. \quad (16)$$

In practice, $S(\alpha)$ contains quantum states whose Bloch vector belongs to a ring parallel to the equator at colatitude α .

The task called *GHZ* in Fig. 8 describes a task where one wishes to generate states from a family of 4-qubit entangled GHZ-type states with varying relative phases. Each class corresponds to

$$S(\mathbf{x}) = \left\{ \frac{1}{\sqrt{2}} (|\mathbf{x}\rangle + e^{i\phi} |\bar{\mathbf{x}}\rangle) \quad \text{s.t.} \quad \phi \sim \mathcal{U}(0, 2\pi) \right\}, \quad (17)$$

where $\mathbf{x}$ is a 4-bit binary string, and $\bar{\mathbf{x}}$ denotes its bitwise complement. We take $\mathbf{x} \in \{0000, 0001, 0010, 0100, 1000, 0011, 1001, 0101\}$ to avoid double-counting equivalent pairs. These GHZ-type states span a structured subspace of the 4-qubit Hilbert space and present a more challenging generative task due to their entanglement and symmetry properties.

Across both tasks, we observe a consistent trend: as the number of target classes $|C|$ increases, the testing loss $\mathcal{L}(0)$ also increases. However, the rate and scale of degradation differ significantly between the two cases. As shown in Fig. 8(a), the error for the 4-qubit GHZ task is an order of magnitude larger than that for the single-qubit equator task at the same class count. This disparity highlights how the structure of the target states impacts model performance under fixed computational resources.

VI. CONCLUSION AND OUTLOOK

The CQDD model introduced in this work represents a significant advancement toward more powerful and ex-

pressive quantum machine learning. By employing a shared set of parameters to generate distinct classes of quantum states, our model eliminates the need to train separate networks for each class. The conditioning mechanism introduces only a minor overhead. Namely, one additional gate on each ancilla qubit, which results in a negligible increase in the circuit depth for current hardware.

We validated the model across four diverse datasets: (i) a set of single-qubit states arranged on a ring in the Bloch sphere, (ii) clusters of states around poles of the Bloch sphere, (iii) two-qubit Bell states with a random relative phase, and (iv) many-body ground states of the transverse-longitudinal field Ising model. These tasks span both abstract and physically motivated distributions, demonstrating the generalisation capabilities and flexibility of the CQDD framework. Benchmarking against an unconditioned model was also performed to highlight the need for a conditioned model and showed a 10 fold increase in performance. Furthermore, an ablation study was conducted, examining the role of key hyperparameters such as the number of denoising steps T , the circuit depth L , and the number of ancilla qubits n_a . Lastly, we look at problem-specific factors such as the number of classes one would wish to generate from.

Despite the strong performance of the CQDD model, several important questions remain open. First, how well does the model generalise to unseen classes? The training stage is performed with a finite number of settings of the angle μ with which the ancilla qubits are initially prepared. It is not clear what would happen when using a trained model by setting the conditioning angles μ to values different than those employed during training. This is especially relevant for continuous or high-dimensional conditioning spaces, where exhaustive training is impractical. We provide preliminary results in Appendix D, but a more thorough investigation is needed.

Second, can the conditioning mechanism be learned end-to-end rather than manually specified? Parametrising and optimising the conditioning jointly with the generative model could lead to more efficient or task-specific encodings. This is akin to learned embeddings in classical conditional generative models such as classifier-free guidance [41] or self-supervised learning [42, 43].

Finally, assessing the model's robustness under realistic noise is essential for near-term implementation. While current results are based on ideal simulations, practical devices introduce decoherence, gate errors, and readout noise. Exploring performance under such imperfections, as well as the potential of noise-aware training or error mitigation, is an important step toward hardware execution.

ACKNOWLEDGMENTS

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Problem	Anc.	RX	RY
2 Equator Rings	11%	6.4%	9.1%
4 Equator Rings	4.4%	4.0%	5.2%
Entanglement	1.8%	1.4%	1.4%

TABLE II. **Comparison of Conditioning Methods.** Values of the final test loss $\mathcal{L}(0)$ for different conditioning approaches: ancilla-based initialisation (Anc.) and label encoding via rotation gates (RX and RY). All experiments for each method use identical model hyperparameters with $n_a = 2$ and consistent random seeds for data generation and training.

use of the computing resources from the Northern Ireland High Performance Computing (NI-HPC) service funded by UK EPSRC Grant No. EP/T022175.

DATA AVAILABILITY

The source code used in the current study will be openly available at [44] with the final accepted version of the paper.

Appendix A: Comparison of Conditioning Methods

Table II presents the values of the final testing loss obtained using various conditioning methods across different multi-class generation tasks. For the 2-class problem, ancilla qubits are initialised to the computational basis states $|00\rangle$ and $|11\rangle$, with conditioning angles set to 0 and π , respectively. In the 4-class case, the ancilla system is initialised to all four computational basis states, with conditioning angles of 0, $\pi/2$, π , and $3\pi/2$. The problem denoted ‘equator rings’ is the task as described in Section VB given by Eq. (15). For the entanglement task, we use the problem as discussed in Section IV C for $n = 2$ qubits.

We omit the results using the RZ rotation gate, as it performed significantly worse in all cases. This underperformance arises because $RZ(\mu)|0\rangle = e^{i\mu/2}|0\rangle$ introduces only a global phase, which has no significant effects on the state. As a result, the conditioning label fails to encode meaningful class information, and the model is unable to distinguish between different input classes. Consequently, the model converges to a distribution that represents an average over all classes. For example, in the two-class equator ring scenario, where each class corresponds to a ring offset from the equator, the RZ conditioning leads our CQDD model to generate a distribution centred around the equator, which was not explicitly part of the training data. This behaviour is reminiscent of the *mode collapse* in classical generative models, where the generator maps multiple inputs to the same output, leading to poor generalisation and low sample diversity [45].

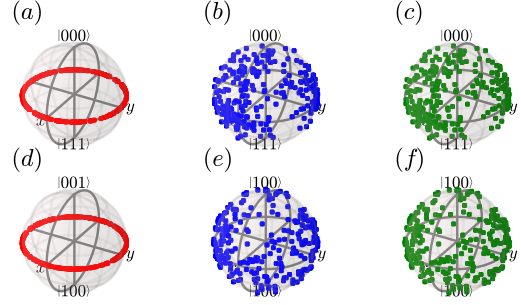


FIG. 9. **Projected Bloch Spheres.** (a-c) Projection of the target, training, and testing GHZ states onto the subspace spanned by $\{|000\rangle, |111\rangle\}$. (d-f) Projection of the target, training, and testing W states onto the subspace spanned by $\{|001\rangle, |100\rangle\}$. All the plotted states have been renormalised.

Appendix B: Entanglement Rings and 3 Qubit Results

Visualising the full entanglement structure of two- or three-qubit quantum states is inherently non-trivial due to the high dimensionality of the Hilbert space. We employ a projected Bloch sphere representation focused on specific subspaces. For the S_Φ class from Eq. (11), we project the state onto the subspace spanned by $\{|00\rangle, |11\rangle\}$. This projection is isomorphic to a single-qubit state of the form $(|0\rangle + e^{i\phi}|1\rangle)/\sqrt{2}$, and hence, we expect the Bloch sphere to exhibit a ring-like structure in the XY -plane.

We extend our approach to $n = 3$ qubits and define two distinct target distributions. One class is S_{GHZ} with states $|\text{GHZ}\rangle = |000\rangle + e^{i\phi}|111\rangle$ up to some normalisation, and the other class is S_W that includes states $|\text{W}\rangle = |001\rangle + |010\rangle + e^{i\phi}|100\rangle$, again up to the appropriate normalisation. As before, we employ a subspace percentage metric to evaluate the generation quality. Take a general 3 qubit state $|\psi\rangle = \sum_i c_i |i\rangle$, then the percentage of overlapping with the subspace of GHZ states is $|c_0|^2 + |c_7|^2$, while for the W states is $|c_1|^2 + |c_2|^2 + |c_4|^2$. The model successfully captures both GHZ and W classes, with roughly 90% of the generated samples lying in the correct subspace. However, the visual structure of the generated data differs significantly from the two-qubit case.

In Fig. 9(a)-(c), we project the S_{GHZ} onto the subspace spanned by $\{|000\rangle, |111\rangle\}$, which is isomorphic to a single qubit. As one can see, even though we are 90% in the right subspace (i.e., the projection into this subspace is faithful) we do not yield a clean ring upon projection from the training and testing. Similarly, the W-class states, which lie in the rank-3 subspace $\{|001\rangle, |010\rangle, |100\rangle\}$, do not yield a clean geometric structure upon projection, as shown in Fig. 9(d)-(e).

Appendix C: Product and Entangled States

Contrary to naive expectations, learning product states is empirically more difficult than learning entangled states. We conjecture that this originates from the structure and effective rank of the states involved. Below, for simplicity, we explore the case of generating a single class of states.

Consider first the two-qubit case. A typical product state with a random local phase ϕ takes the form

$$S_{\text{PS}} = \{(|0\rangle + e^{i\phi}|1\rangle)^{\otimes 2} \text{ s.t. } \phi \sim \mathcal{U}(0, 2\pi)\}, \quad (\text{C1})$$

whereas an entangled state with the same phase dependence can be written as

$$S_{\text{Ent}} = \left\{ \frac{|00\rangle + e^{i\phi}|11\rangle}{\sqrt{2}} \text{ s.t. } \phi \sim \mathcal{U}(0, 2\pi) \right\}. \quad (\text{C2})$$

Expanding the product state, we see $(|0\rangle + e^{i\phi}|1\rangle)^{\otimes 2} = |00\rangle + e^{i\phi}(|01\rangle + |10\rangle) + e^{2i\phi}|11\rangle$ up to some normalisation. The product state features phase contributions of both ϕ and 2ϕ , while the entangled state depends only on a single phase. In this sense, the entangled state resides in a lower-dimensional manifold and is effectively governed by a single-qubit parameter. By contrast, the product state exhibits full rank over the Hilbert space and introduces multiple independent phase components, thus increasing its representational complexity.

This effect becomes even more pronounced for three qubits. The corresponding set of product states is

$$S_{\text{PS}} = \{(|0\rangle + e^{i\phi}|1\rangle)^{\otimes 3} \text{ s.t. } \phi \sim \mathcal{U}(0, 2\pi)\}, \quad (\text{C3})$$

which expands as $|000\rangle + e^{i\phi}(|001\rangle + |010\rangle + |100\rangle) + e^{2i\phi}(|011\rangle + |110\rangle + |101\rangle) + e^{3i\phi}|111\rangle$ for a single term, up to some normalisation. The entangled state forms a set of GHZ states,

$$S_{\text{Ent}} = \left\{ \frac{|000\rangle + e^{i\phi}|111\rangle}{\sqrt{2}} \text{ s.t. } \phi \sim \mathcal{U}(0, 2\pi) \right\}. \quad (\text{C4})$$

Again, the product state distributes the phase across many amplitudes, while the entangled state retains a simpler structure. Empirical evidence supports this structural interpretation. For two qubits with $n_a = 1$ ancilla qubits, $T = 20$ diffusion steps and $L = 10$ circuit depth, the final test loss for the product state is approximately 2.2%, compared to the 0.51% for the entangled state (roughly 4.4 times larger). For three qubits under the same settings, the loss for the product state is 17%, compared to 1.8% for the entangled state (roughly 9.5 times larger). These results suggest that higher-rank states with a more distributed phase structure are indeed harder to learn for the CQDD model.

Appendix D: Unseen Values of Conditioning Angle μ

We assess the model's ability to generalise to conditioning angles μ not encountered during the training.

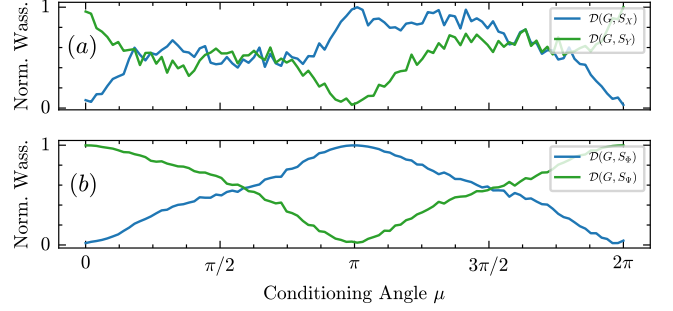


FIG. 10. **Generation Distance.** Normalised Wasserstein distance between $G(\mu)/\max G(\mu)$ and the trained target distributions. (a) Distances to S_X (blue) and S_Y (green), where S_X corresponds to $\mu = 0$ and S_Y to $\mu = \pi$. (b) Distances to S_Φ (blue) and S_Ψ (green) under analogous conditioning.

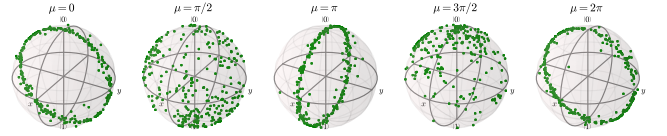


FIG. 11. **Intermediate Conditioning.** Bloch-sphere visualisation of $G(\mu)$ for selected values of μ . Each Bloch sphere corresponds to a different conditioning angle. For unseen angles $\mu = \pi/2$ and $3\pi/2$, the generated states lack the structured rings of the target sets.

Specifically, we sweep $\mu \in [0, 2\pi]$ in uniform steps and compute the normalised Wasserstein distance between the generated distribution $G(\mu)$ and its corresponding target set. Two tasks are considered, (i) generation of the state classes S_X and S_Y defined in Eqs. (6) and (7), and (ii) generation of the entangled state classes S_Φ and S_Ψ defined in Eqs. (11) and (12). In both tasks, $\mu = 0$ and $\mu = \pi$ correspond to the two target classes and both models use $T = 20$, $L = 12$, $n_a = 2$, and $N = 125$.

For the rings task of Fig. 10(a), the distance to S_X increases with μ up to a maximum at $\mu = \pi$, while the distance to S_Y follows a complementary trend, such that $\mathcal{D}(G, S_X) + \mathcal{D}(G, S_Y)$ remains approximately constant. This indicates that the model maintains a consistent trade-off between the two classes, producing a distribution close to the intended target as long as μ is near the desired angle. However, Bloch sphere visualisations, as given in Fig. 11, reveal that for unseen intermediate angles ($\mu = \pi/2, 3\pi/2$) the generated states exhibit no resemblance to either target distribution, forming clusters of random points rather than well-defined rings in different planes.

A smoother interpolation is observed for the entanglement task of Fig. 10(b), with the crossover of the curves at $\mu \approx \pi/2$ and $\mu \approx 3\pi/2$. At these points, the generated states are nearly equidistant from both target classes, and occupy both subspaces $\{|00\rangle, |11\rangle\}$ and $\{|01\rangle, |10\rangle\}$ with a 60/40 split favouring $\{|00\rangle, |11\rangle\}$. These intermedi-

ate states, absent from the training set, suggest that the model can interpolate between entangled targets, albeit without preserving the fine structure of either distribution.

Appendix E: Architectural and Simulation Details

Table III shows the model hyperparameters used for each of the problems discussed in the main text.

The simulation of the CQDD model is carried out using the Python library `TensorCircuit` [46], while Bloch sphere visualisations are generated with the assistance of `QuTiP` [47]. Computation of the Wasserstein distance is performed using the Python Optimal Transport library `POT` [48].

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Problem	$ \mathcal{C} $	n	n_a	L	T	N	δ_t	Distance	Performance
Planar Rings (IV A)	3	1	2	15	20	1000	$0.005t^2$	WASS	4.3%
Polar Points (IV B)	6	1	2	12	20	500	$0.15t$	MMD	1.5%
Entanglement Structure (IV C)	2	2	2	12	20	125	$0.01t^2$	WASS	2%
Many Body Phases (IV D)	2	4	2	12	30	100	0.1 to 2	MMD	3.7%

TABLE III. **Model Hyperparameters and Performance.** Summary of the CQDD model hyperparameters and its performance across various generative learning tasks. $|\mathcal{C}|$ indicates the number of different classes the model wishes to generate from. The number of data qubits is n , and n_a denotes the number of ancilla qubits. L is the depth of each individual denoising circuit $\tilde{U}$, and T is the number of diffusion steps. N represents the number of samples used during training and testing. δ_t refers to the noise schedule. The notation x to y implies a linearly spaced set of length T starting for x and ending at y . Distance denotes which metric we use, either maximum mean discrepancy (MMD) or Wasserstein (WASS). Performance is reported as the final test value of the loss function $\mathcal{L}(0)$.

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